



The Evolution of Biomechanical Analysis: From Lab-Based Systems to AI-Driven Markerless Tools

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Abstract—Biomechanical analysis is widely used in sports to evaluate human movement, improve performance, and reduce injury risk. Traditional biomechanical analysis mainly uses laboratory-based optical motion capture systems, which provide highly accurate measurements but require controlled laboratory setups, expensive equipment, and technical experts. Due to these limitations, their use in field-based sports training is limited. Recent advancements in wearable sensors and computer vision have introduced alternative approaches for motion analysis. Inertial Measurement Unit (IMU) systems provide portable solutions for field-based monitoring, while Machine learning based markerless motion capture systems allow movement analysis using standard video cameras. This literature review examines the evolution of biomechanical analysis technologies, focusing on marker-based optical systems, IMU-based wearable sensors, and markerless motion capture methods. The review explore their advantages, limitations, and suitability for analyzing high-speed movements such as cricket bowling. Findings suggest that although marker-based systems remain the most accurate, markerless approaches show strong potential for developing accessible and cost-effective biomechanical analysis tools for field-based sports environments.

Index Terms—biomechanical analysis, marker-based motion capture, markerless motion capture, Cricket biomechanics, pose estimation, computer vision, Machine learning

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I. INTRODUCTION

Biomechanical analysis is the study of human movements often used in sports, research and rehabilitation to improve performance and prevent injuries. It involves assessing motion, forces and muscle activity to identify how the body moves efficiently. Accurate biomechanical analysis has the potential to provide accurate coaching feedback, identify injury risks and improve athlete monitoring. For example, analyzing a fast bowler in cricket allows coaches to identify inefficient or risky techniques and provide data-driven feedback to enhance performance and reduce injury risk.

Traditional biomechanical analysis requires specialized laboratory based tools which are expensive, require controlled environments and are not easy to use in outdoor field setting. These systems provide very detailed measurements of human motion. However, their limitations reduce the opportunities for athletes to access these systems in field environments.

With the recent advancement of technology, biomechanical analysis has become more accessible, efficient and cost effective. AI and ML have transformed the field with markerless motion tools that use smartphone cameras and standard RGB videos to assess athletes' movements. These tools remove the need for expensive tools and make biomechanical analysis

available to a wider community to analyze player performance. However, their specific suitability and accuracy for high speed sports movements remain to be fully established.

Therefore it is essential to continue research and development to enhance these AI/ML driven models, mitigate their limitations and provide more reliable and accurate results that can easily be interpreted and applied in cricket bowling.

This narrative literature review focuses on the evolution of biomechanical analysis tools in cricket bowling, from traditional laboratory-based systems to modern AI/ML-based markerless motion capture. It reviews key concepts, definitions, advantages and limitations of different approaches and describes the practical implications. By exploring these tools, the review identifies gaps in current research and highlights the need for cost friendly, accurate, and easy to use solutions that can be applied in field-based sports environments. Improving access to reliable biomechanical analysis can ultimately improve athletic performance, support data driven coaching, and make training easier and more effective for athletes at all levels.

II. BACKGROUND

Biomechanics plays an important role in understanding and improving athletic performance, especially in sports involving difficult movements like cricket. Bowling in cricket is considered one of the most biomechanically demanding athletic actions, as it requires rapid, coordinated movements involving multiple body parts. Accurate evaluation of key parameters such as joint angles (elbow and knee flexion-extension), segment velocities (wrist and arm), stride lengths and durations, and Center of mass trajectories is important for performance optimization and injury prevention [1] [2].

Traditionally, most of the biomechanical analysis has been done using marker-based motion capture systems, high speed video analysis and force platforms. However, these systems are expensive, require controlled laboratory environments and lack portability. Therefore, even though they offer high accuracy, their use is limited at the grassroots level. Additionally, since these systems require technical expertise, most coaches and athletes, particularly in developing regions lack access to them [3].

Recent advancements in Machine learning (ML) and pose estimation offer new approaches for accessible, cost effective biomechanical analysis. After extracting 2D or 3D joint key points using pose estimation tools and integrating them with ML models, the data can be used to classify bowling types, analyze movements, and provide accurate feedback. Overall, these technologies provide a foundation for creating low-cost, portable, and scalable biomechanical tools suitable for environments beyond professional labs [3].

III. REVIEW METHODOLOGY

A structured literature search was conducted using Google Scholar between January 2000 and January 2025. Keywords used includes "cricket bowling biomechanics," "markerless motion capture tools," "pose estimation validation," "AI and

ML based pose estimation tools,” “inertial measurement units,” “Traditional biomechanical analysis tools”

Studies were included if they met at least one of the following criteria: (1) peer-reviewed journal or conference papers evaluating biomechanical motion analysis tools in sports contexts, (2) studies validating motion capture technologies such as optical marker-based systems, IMU sensors, or markerless pose estimation methods, (3) research focusing on high-speed human motion analysis relevant to sports such as cricket bowling, running, or throwing, or (4) studies reporting quantitative accuracy metrics such as joint angle errors, positional errors, or root mean square error values. This inclusive approach was used to capture the wide range of technologies used in biomechanical motion analysis and to understand their advantages and limitations.

Studies were excluded if they were non-academic sources such as blogs or reports, lacked sufficient methodological description, or focused solely on activity classification without addressing biomechanical motion analysis. The remaining studies were thoroughly reviewed to understand and analyze the evolution of biomechanical motion analysis tools. The selected studies provide the understanding of current capabilities, limitations and future directions in biomechanical human motion analysis.

IV. CRITICAL REVIEW OF LITERATURE

in sports biomechanics, motion capture is basically done using 3 types of technologies which are optical marker based systems, Inertial measurement unit (IMU) systems, and Markerless systems using computer vision. Each has their own advantages and limitations in terms of accuracy, cost and real world usability. Understanding these differences are important to analyze the current limitations and future directions in capturing motions of players more accurately.

A. Optical marker based Systems

optical marker based systems are widely used laboratory-based tools that used to track human movements in three dimensions with high accuracy in space and time. they use multiple high speed cameras with small reflective markers to capture body movements. The cameras record the markers from different angles and then the system calculates their exact position in 3D space. Because of the high accuracy, these tools are considered standard methods in biomechanical research. [4]

Recent studies show that these tools provide highly accurate kinetic measurement and because of that they are widely used in sports biomechanical researches. In cricket bowling, these tools can be used to measure key parameters such as Bowling stride length, Shoulder rotation during delivery, Elbow angle, arm speed and Hip-shoulder coordination. According to Colyer et al., marker-based optical motion capture systems remain one of the most reliable methods for capture human movements in sports and rehabilitation research as they provide detailed measurements of joint angles and human movements. [5]

Among widely used optical motion capture systems, Vicon and OptiTrack known for their high accuracy in capturing human motion. Vicon systems are widely recognized for their accuracy and they are mostly used as a reference system in biomechanical research and clinical gait analysis. [6] On the other hand, studies shows that OptiTrack systems provide high accuracy measurements when proper calibration are applied.

OptiTrack motion capture systems are crucial in biomechanical analysis, capturing high fidelity movements needed for clinical research and sports science. The systems capture movement in three-dimensional space through the use of optical cameras and retroreflective markers, capturing movement through stereophotogrammetry, a technique of computing coordinates from multiple photographic perspectives. Precise, dependable, and affordable, OptiTrack systems have extensive use in a range of biomechanical studies including gait analysis and the evaluation of athletic movement.

Although OptiTrack tends to be slightly less precise in very dynamic movements than systems such as Vicon, it provides adequate precision for most clinical and research purposes at a lower price. This makes it an attractive choice in cost-sensitive research environments. Literature reviews, however, indicate a common problem with the use of OptiTrack in research: the under-documentation of important methodological information such as camera parameters and marker placement. This monitoring can have an effect on the reproducibility of outcomes and researchers' capacity to accurately replicate studies.

In practical application, OptiTrack performs effectively in accurate motion analysis in a variety of studies that range from rehabilitation through optimizing sports performance. It provides real-time feedback, which is invaluable in the context of refining interventions and monitoring treatment effectiveness.

Despite its extensive application and growing research evidence, OptiTrack methods require further validation and standardization. Experimental configurations need to be better described to facilitate more systematic comparative analyses and results generalization. In general, OptiTrack motion capture technology is a convergence of precision, practicality, and cost-effectiveness that makes them a good piece of equipment to be used for biomechanical studies. As technology continues to advance, OptiTrack is poised to become even more pivotal in the field of biomechanics with their application subject to strict documentation to ensure credibility and applicability. [7]

Vicon is renowned for its cutting-edge motion capture systems, which form the focal point of biomechanical analysis within clinical and research settings. They utilize high-precision cameras and reflective markers to capture accurate three-dimensional human body movement. This allows for advanced analysis of human kinetics and kinematics, which are paramount across fields ranging from rehabilitation to sports science.

Vicon systems employ a number of configurable marker sets, such as the full Plug-in-Gait, for which accurate anthropometric data are required to produce good motion modeling. The systems may utilize between six cameras for basic motion

capture and over a dozen for higher-level activities in order to provide complete spatial coverage and accuracy in data.

They perfectly complement force platforms and electromyography (EMG) to provide a full understanding of biomechanical performance by combining the kinematic data with the kinetic forces and muscular activity. Vicon's software provides complex data processing with automatic labeling and gap filling that enhances the motion analysis robustness.

In clinical environments, Vicon is used to assess gait disorders and monitor rehabilitation, adding empirical support to treatment decisions. In research, it offers informative data on athletic performance and ergonomic design, allowing for the optimization of movement and prevention of injury. In summary, Vicon's motion capture technology is an essential asset to precise biomechanical analysis with adjustable setup and in-depth data processing capability supporting a wide range of applications both in research and clinical domains. [8]

1) *Accuracy and precision:* One of the main advantages of optional motion capture systems is their accuracy and precision. Validation studies have shown that Vicon systems have demonstrated static mean absolute errors of 0.15 mm and dynamic errors below 2 mm. Sometimes it can be as low as 0.3 mm depending on the marker size and sampling rate. [9] Similarly, OptiTrack systems usually have error below 0.2mm and in smaller areas it can be below 0.1mm.

2) *Limitations:* Even though optical motion capture provide accurate results, they have several limitations as well. One of the main limitation is that these systems are expensive and need specialized lab environment. In addition to that, they need careful marker placement, proper calibration and trained people to operate them. However despite these challenges, these system remains most reliable tools for biomechanical analysis. [10]

B. Inertial measurement unit (IMU) systems

IMU systems use attachable sensors to capture the movements of athletes. these systems are used for the measurement of static and dynamic states of the athlete body. some of the most important parameters that IMU systems measure are linear acceleration, spatial position, angles between body parts, angular velocity and displacement. [11]

Recent studies show that IMU based motion capture systems are widely used in sports biomechanics because they are portable and can be used field based environments. in sports like cricket, IMU sensors are used to measure movement patterns by attaching sensors to body parts like the arm, trunk and legs. These sensors allow researchers to analyze parameters such as joint rotation, body acceleration and angular velocity during the delivery phase. According to Camomilla et al., wearable inertial sensors provide useful biomechanical measurements in sports and rehabilitation environments where laboratory systems are not practical. [12]

there are several IMU based systems are available in the industry. Systems such as Xsens MVn, APDM Opal, Kinexon

and MotusBASEBALL are commonly used IMU based systems to capture human motion and measure joint kinematics using sensor fusion algorithms that combines accelerometer, gyroscope and magnetometer data. [13]

Kinexon is a wearable sensor-based motion tracking system that uses ultra-wideband (UWB) technology to monitor player movements, acceleration, and workload in real-time. It provides affordable and user-friendly solutions for sports analysis, specifically in indoor games like volleyball and handball to track physical demands and optimize training regiments.

One of the key advantages of kinexon is that it is easily accessible for non technical users like coaches and athletes. The system consists of light weight sensors that can be worn by athletes along with an intuitive software that processes data automatically. Apart from that kinexon is really affordable compared to high end technical tools which is also a great advantage. While it provides great accuracy for speed and acceleration measurements, it struggles with high Deceleration values. Additionally, sensor placement also can affect the accuracy of results. [14]

MotusBASEBALL is a wearable inertial measurement unit (IMU) sensor designed to analyze the biomechanics of a pitcher's throwing motion. It offers a user-friendly, real-time way to monitor arm slot, shoulder rotation, arm speed, and stress levels of pitcher. One of the key advantages of MotusBASEBALL is its affordability compared to high-end motion capture systems like OptiTrack or Vicon. The sensor fits into a compression sleeve on the arm, and connects to a mobile app, removing the need for lab-based motion tracking. The app provides instant feedback, allowing users to track performance over time without requiring in-depth biomechanical knowledge.

Its accuracy in assessing arm speed and stress is less reliable although it has good correlations with traditional motion capture systems for parameters like arm slot and shoulder rotation. The study found that MotusBASEBALL underestimated arm speed by around 80% compared to motion capture, and stress measurements were approximately 39% lower than those from high-precision systems. Additionally, because IMU sensors rely on accelerometers and gyroscopes, they can introduce errors when measuring movements in multiple planes, limiting their precision in advanced biomechanical research. [15]

Overall, wearable devices have revolutionized biomechanical analysis by providing real time monitoring. But still there are some limitations as well. Such as the accuracy of these devices depends on sensor placement, environment factors and type of movement. Apart from that, the cost of some advanced devices can be high which prevents non-elite athletes from using them. And some users might feel uncomfortable wearing the device. So as the technology advances these tools are expected to be more accurate, affordable and user friendly.

1) *Accuracy and precision:* According to the studies, IMU systems are generally show lower positional accuracy compared to optical motion capture systems. for example a study has done to compare Xsens IMU system with optoelectronic motion capture system and reported root means square error

from about 1.2° during simple movements to about 2.8° during complex movements. this shows that accuracy can be affected by complexity and duration of the movement tasks. [16]

Other validation studies also show that IMU systems can provide reliable measurements of joint kinematics in dynamic sports movements when proper calibration procedures are applied. [17]

2) *Limitations:* One common limitation is sensor drift, where small measurement errors accumulate over time and affect the accuracy of position estimation. In addition, the accuracy of measurements can be affected by incorrect sensor placement and magnetic interference in the environment. [16]

Another Limitation is that players face difficulties in playing while wearing sensor. they might get distracted from them and this can be affected to their performance. In addition to that IMU systems mainly estimate orientation and acceleration, which makes it difficult to obtain highly accurate joint position data compared to optical motion capture systems. Because of that, IMU systems are often used for field based motion analysis but are still validated against optical systems in laboratory environments. [12]

C. Markerless systems using computer vision

Markerless motion capture system capture human motion using standard videos without the need for markers. these systems track skeleton points in 2D and 3D to reconstruct joint kinematics using computer vision and machine learning. [18]

In recent past, there has been several markerless systems developed. DeepCutLab is one of the widely used markerless system that provide highly accurate results. studies shows that it provide human level accuracy in detecting keypoints and there are errors of only a few pixels when there is sufficient training data. [18]

With DeepLabCut, users are now able to analyze motion without the traditional use of motion capture markers. It serves as an utterly revolutionary open sourced tool box ideal for neuroscientists and mechanics, It is also frequently used in plenty of fields, including ethology, sports science, and rehabilitation. DeepLabCut offers the advanced capability of deep learning, especially through the use of transfer learning for accurate pose estimation. Instead of using sensors or markers to track body movement, DeepLabCut enables researchers to build custom neural networks that detect and track body parts, whether the subject is a human, an animal, or a robotic system.

Flexibility is one of DeepLabCut's strongest advantages. Researchers are not restricted to accepting pre-trained models that can offer limited results. Instead, they can build their own by simply marking a few frames on a video, DeepLabCut will then generalize the labeled features across the entire video. This powerful feature enables unprecedented biomechanical analysis, especially in research cases for setting where the placement of motion markers is non-invasive and could potentially get in the way of natural movement.

The use of DeepLabCut in neuroscience and behavioral studies has been quite remarkable. It has been utilized for

gait analysis, limb movement coordination, and even small motor skills in small animals like mice and insects. It gives more researchers the ability to track movements with high precision and acts as an alternative for expensive lab-based motion capture tools in biomechanics.

Piloting over the rest of the features is the fact that DeepLabCut is hardware independent. DeepLabCut works without specialized cameras or expensive motion tracking devices. Rather, it works with the footage captured by normal video cameras. This makes it a low-cost, easily scalable option for motion analysis. Because of this feature, adoption has increased in the clinical research field where adolescents for movement disorders, rehabilitation, and physical therapy progress are Analyzed.

Some of the areas in which DeepLabCut has been applied include:

- 1) Sports science – studying the posture and movements of an athlete and their efficiency
- 2) Rehabilitation and physical therapy – Improvement and tracking of progress of patients
- 3) Neuroscience and animal studies – fine motor skill activities in animals during controlled experiments
- 4) Ergonomics and occupational health – evaluation of worker's movements to prevent injuries

DeepLabCut's innovative approach is without the use of motion-capture labs, which helps democratize biomechanical research. [18]

In addition, OpenPose is one of the famous markerless motion capture system that detects human skeleton points in real time using RGB video. this tool is widely used in sports analysis and rehabilitation to identify multiple body joints including arms, legs and facial landmarks. [19] However, its accuracy can be affected by fast movements and external factors such as lighting conditions and occlusions.

This tool single-handedly advanced the technique for analyzing a single image or frame from a video and identifying critical areas on a person's body, face, hands, and feet with a novel bottom-up technique. Unlike other systems which require markers to be placed on a person's body for movement to be captured, OpenPose can function with a standard camera and does not require any additional markers or equipment.

It utilizes a deep learning model with an additional part affinity field (PAF) to link detected body sections to the corresponding person. This revolutionized the system by increasing the accuracy of tracking multiple people at once while still ensuring simultaneous processing of the video imagery. Nowadays OpenPose is frequently used in sports analytics, rehabilitation, ergonomics, and gaming.

OpenPose's weaknesses involve limitations on occluded or low-light surroundings where accuracy of pose estimation drops significantly. But, like many pose estimation systems, it is critical to computer vision-based biomechanics research. [20]

MediaPipe is another powerful open-source framework that achieves effective real-time pose estimation. This system was introduced as a lightweight and cross-platform solution for

mobile and web applications. MediaPipe, unlike OpenPose that runs primarily on GPU, is designed for low-power devices. This includes smartphones and other embedded systems.

MediaPipe Pose utilizes machine learning models that have been trained with large-scale human pose datasets. The pipeline model utilizes three main stages:

- 1) Person detection - This step detects the human subjects within the given frame.
- 2) Pose estimation - This step extracts skeletal key points of the detected person.
- 3) Post-processing - The detected key points are normalized while pose refinement is done to achieve optimal accuracy.

OpenPose is more accurate in a multi-person setting, but MediaPipe is faster. It is significantly more lightweight which allows for real-time tracking without the aid of a GPU. MediaPipe has a wider range of applications including fitness tracking, augmented reality (AR), and motion and gesture-based systems. Although, OpenPose is still more accurate in complex multi-person scenarios. [21]

OpenCap is another markerless motion capture system that is being widely used to estimate three dimensional kinematics and musculoskeletal dynamic from video data. Validation studies show that OpenCap can estimate measures such as joint moments and muscle activations accurately while being faster and less expensive than marker-based motion capture systems. [22]

PoseNet is a deep learning model aimed to achieve human pose estimation in real time, capturing some key body parts from photos or video clips. PoseNet tracking systems are infringing skeletal tracking systems which do not need specialized equipment or marker-based motion functions. Mixed biomechanical analysis aids which depend on external biometrics use sensors are unlike PoseNet. PoseNet tracking system captures images through RGB and processes these images to estimate possible positions of human joints by employing convolutional neural networks.

PoseNet supports a two-step pipeline for pose estimation:

- 1) Feature Extraction – Processes like MobileNet and ResNet run deep CNNs on the given image to extract essential spatial features.
- 2) Pose Keypoint Estimation – Captures features that generate probability heatmaps and offset vectors to localize essential joints in head, shoulders, elbows, or knees.

Advantage range across all computing environments is fundamental use of PoseNet features. Models like OpenPose need GPUs for effortless multi-live tracking biometrics recognition. PoseNet changes the game by functioning perfectly with mobile devices and web browsers making it fit for use in remote assistance, rehabilitation, and human-computer interaction and augmented reality (AR).

Although PoseNet delivers a solution for estimating human positions with great speed and efficiency, its accuracy is poor in complicated situations like tracking multiple people or occluded body parts. It is not as precise as marker-enabled

motion capture systems and can also fail in low-light or active scenarios. Nonetheless, the lightweight structure and real-time features make it an important component of biomechanical analysis in the field, where accessibility and costs are important factors. [23]

1) Accuracy and Precision: The previous studies have shown mixed results for the accuracy and precision of markerless motion capture systems. The studies comparing DeepLabCut and OpenPose with marker-based systems show root mean square errors of approximately 5° to 10° for hip, knee and ankle joints during running [24]. This suggests that these systems can provide reasonably accurate measurements.

Another study has reported median joint localization errors of pose estimation algorithms ranging from 9mm to 22mm when detecting limb anatomical landmarks. [25] However, these systems provide lower precision than laboratory-based optical motion capture systems.

2) Limitations: Even though markerless motion capture tools are low cost and easy to use, they face issues with accuracy and precision. One of the main challenges is that their accuracy can be affected by external factors such as camera angles, lighting conditions and occlusion of body parts. [26]

Another limitation is that many markerless systems rely on two-dimensional videos which can lead to errors when measuring three-dimensional joint kinematics. As a result, these tools may not achieve the same level of precision compared to marker-based optical motion capture systems.

THE SHIFT TOWARD MARKERLESS MOTION ANALYSIS

Recent advances in computer vision and artificial intelligence (AI) have enabled markerless motion capture, reducing the need for costly lab-based equipment. Colyer et al. (2018) encapsulate the evolution of vision-based motion tracking systems, noting that machine learning and deep learning algorithms can now identify and analyze human movement with or without invasive body-worn markers. [5] These advances have opened up new possibilities for low-cost, real-time biomechanical analysis outside the laboratory. Their contribution highlights that while marker-based systems remain the gold standard for accuracy, marker-based systems are constrained by expense, time, and natural movement capacity limitations.

In contrast, markerless approaches, patterned after approaches in the video game and film industries, employ computer vision to eliminate markers from being required to recover core biomechanical parameters and thereby allow greater universal usability and accessibility. [27] By removing the need for physical markers, these systems enable greater flexibility, allowing athletes to perform in natural environments without the need of wearable sensors. This shift significantly expands the accessibility of biomechanical analysis, enabling applications beyond elite sports laboratories, including schools, amateur clubs, and rehabilitation centers.

Furthermore, markerless motion capture systems can measure movements using the data collected from RGB cameras or even smartphone footage making them highly portable.

Advanced deep learning models such as convolutional neural networks (CNNs) and pose estimation frameworks like OpenPose, DeepLabCut, and MediaPipe can use to track multiple body joints in real time, detect rapid movements, and even analyze multi-person interactions at the same time. These capabilities make these systems highly reliable for sports like cricket, where bowling involves rapid and complex movements.

Even though there are many advantages of markerless motion capture systems, the challenges remain. one of the main challenges is that the accuracy of these systems can highly affected by external factors like camera placement, lighting and speed of the movement. However, many studies show that markerless systems are getting closer to the accuracy of marker-based system with the improvement of technology.

ACCURACY AND CHALLENGES IN FIELD-BASED MOTION ANALYSIS

Probably the most significant consideration in applying markerless motion analysis to everyday settings is the ability to reach accuracy levels comparable to laboratory-based systems. Colyer et al. (2018) accept that while AI-based vision-based methods have shown promise, their accuracy could be influenced by environmental conditions, the placement of the camera, and the character of the movement being recorded. [5]

Further validation against gold-standard motion capture systems must be conducted to establish their reliability in different applications. Additionally, non-specialist users, such as coaches, athletes, and physical therapists, must be capable of reading and utilizing the analysis effectively. This generates the need for intuitive interfaces, data processing automation, and actionable information that does not require a high degree of biomechanical expertise.

V. CRITICAL ANALYSIS OF WORK

As mentioned above, each category has its own advantages and limitations. Lab-based tools are considered the gold standard due to their high accuracy; however, they are costly and require controlled lab environments, which affects practical implementation. This raises concerns about accessibility for non-technical users, such as coaches and athletes.

As a solution, field-based tools have been developed over the past few years, making biomechanical analysis more accessible to a wider audience. However, these tools face limitations in accuracy due to factors like lighting, environmental conditions, and camera placement.

To meet these requirements, users now use AI/ML based tools for Biomechanical analysis. These tools allow posture and movement tracking in real time and do not require markers or other expensive equipment. Despite overcoming some of the aforementioned concerns, their accuracy is still dependent on the training data, the environment, and the computing power available. The opacity resulting from these new models also

Category	Lab-based Tools	IMU-Based Wearable Systems	AI/ML Markerless Systems
Cost	High	Moderate	Low to Moderate
Accuracy	Very High	Moderate	Depends on training data and environment
Ease of Use	Requires special equipment and experts	Portable, moderate setup	Requires technical knowledge
2D/3D Availability	Mostly 3D	Mostly 3D	Mostly 2D
Portability	Not portable	Highly portable	Highly portable
Real-time Monitoring	Yes	Limited	Yes
Suitability for bowling analysis	Excellent	Good	Promising

TABLE I
COMPARISON OF DIFFERENT BIOMECHANICAL ANALYSIS TOOLS

makes them more complicated, which can be a disadvantage for users with no technical background.

A major issue for these newly developed biomechanical analysis tools is that users are limited to practicality or accuracy rather than having a middle ground, which makes the situation worse. This makes it clear that a single integrated solution that possess the positive factors of all the three systems is needed. Subsequent work should aim at improving accuracy in real time while making the tools affordable and simple, expanding beyond just computer trained experts to coaches, athletes, and trainers as well.

VI. RESEARCH GAP AND OPEN CHALLENGES

Human movement can be measured with extreme accuracy using conventional biomechanical analysis methods such as optical marker-based motion capture systems. However in order to operate these systems require specialized laboratory environments, expensive equipment and skilled technical staff. Due to these constraints, such systems are rarely accessible to coaches and athletes during regular training sessions. As a result biomechanical analysis with laboratory methods is frequently restricted to controlled experimental environmental settings rather than actual sports context

By offering portable and field-based motion analysis solutions, wearable sensor technologies like IMU- based system try to overcome this constraints. Even though these systems enable data collection outside the laboratory, they still face several challenges. The accuracy of the IMU system heavily depends on placement of the sensor, calibration procedure , and environmental conditions such as magnetic interference. Furthermore, measurements inaccuracy and sensor drift can build overtime, making long duration motion analysis less reliable. Additionally , wearable sensors may give athletes discomfort or interfere with their natural athletic movement, which could impair their performance in actual training or competitive situation.

At present, advances in computer vision and machine learning have enabled markerless motion capture approaches that estimate human pose directly from video data. These systems offer several positive aspects, comprising lower cost, requirements of minimal equipments and the ability to analyze movements using standard cameras. Conversely, present markerless approaches still face important limitations. Their accuracy can be affected by lighting conditions, camera angles, occlusions and fast athletic movements. Moreover, many existing systems primarily rely on two-dimensional joint kinematics need for detailed biomechanical analysis.

Another significant limitation identified in existing research is that most computer vision based studies in cricket mainly concentrate on action recognition tasks. These studies typically focus on detecting different bowling types, identifying illegal bowling actions, or classifying player movements. Although such tasks are valuable for automated video analysis, they do not provide detailed biomechanical measurements such as joint angles, stride length, angular velocity or release mechanics. These parameters are essential for improving player performance, preventing injuries and coaching feedback. However, current research does not fully address the need for practical biomechanical measurement tools that can be used directly in training environments.

Moreover, many existing markerless motion analysis systems require stable multi-camera setups, high computational resources, or GPU-based processing pipelines. These requirements minimize their practicality for real-time use by coaches, athletes, and sports analysts who may not have access to advanced computing infrastructure. In addition, the outputs of many AI-based systems are often difficult for non-technical users to interpret, limiting their usefulness in real coaching scenarios.

Another noticeable gap is the lack of integrated solutions that balance accuracy, affordability, and usability. New technologies tend to prioritize one aspect over the others. Laboratory-based systems offer high accuracy but low accessibility, since markerless systems provide accessibility but lower reliability for detailed biomechanical measurements. This trade-off highlights the absence of a practical system that combines the strengths of laboratory-grade accuracy with the portability and affordability required for real-world sports environments.

Consequently, there is a clear research gap in the development of real-time, low-cost, and accessible biomechanical analysis systems especially designed for cricket fast bowling. Such a system should be capable of accurately estimating key biomechanical parameters such as joint angles, stride length, body alignment, and delivery mechanics using simple camera setups while providing interpretable feedback for coaches and players. Addressing this gap would significantly improve the ability of athletes and trainers to analyze bowling techniques during training sessions, enabling data-driven performance improvement and injury prevention outside traditional laboratory settings.

VII. DISCUSSION AND IMPLICATIONS

This narrative literature explores the clear technology of evolution in biomechanical motion analysis, moving from optical marker based motion capture systems toward more accessible markerless motion capture systems. In the beginning, optical marker based systems such as Vicon and OptiTrack have been considered the gold standard for biomechanical analysis because of their high accuracy. These systems provide detailed measurement of human movements which are really important for understanding athletics performance. However, their cost and need of specialized laboratory environments significantly limit their accessibility in field based environments.

Because of these limitations, wearable technologies such as IMU systems have been introduced as portable alternatives for analyzing human movements. These systems allow users to use them in field-based environments, allowing athletes to be monitored in fields rather than in controlled environments. However their accuracy can be affected by sensor drift, magnetic interference, and errors caused by incorrect sensor placement. In addition, wearable sensors may interfere with natural movement patterns or create discomfort for athletes, particularly during high-speed and dynamic movements such as cricket bowling.

With the advancement in computer vision and machine learning, new studies have introduced markerless motion capture systems to analyze human movements. These systems have mitigated most of the limitations of marker based and wearable sensors by allowing motion analysis to be performed using standard video cameras. Modern pose estimation frameworks such as OpenPose, MediaPipe, DeepLabCut, and PoseNet demonstrate the potential of machine learning to automatically detect skeletal keypoints and reconstruct human motion. This shift toward vision-based motion analysis represents a major trend in the field, driven by rapid progress in artificial intelligence, deep learning, and large-scale training datasets.

However, markerless motion capture systems still face several limitations such as their accuracy is highly dependent on environmental conditions such as lighting, camera angles, background, and occlusion of body parts. Fast and highly dynamic movements can further reduce pose estimation accuracy. This specially affects the sports like cricket which has fast and rapid movements. In addition, many markerless systems rely mainly on two-dimensional pose estimation, which may not provide the precise three-dimensional joint kinematics required for detailed biomechanical evaluation. These limitations show the need for improved reliable biomechanical analysis systems that can deliver accurate biomechanical measurements in real-world environments.

From a practical aspect, the growing availability of machine learning-based pose estimation tools has important implications for sports science and coaching. Affordable markerless systems could enable coaches, athletes, and experts to perform biomechanical assessments without using expensive laboratory facilities. This would make biomechanical feedback more accessible during regular training sessions, allowing athletes to

continuously monitor and improve their techniques. In addition, coaches can provide detailed driven feedback to athlete with clear and accurate data rather than guessing. In cricket, such systems could help identify wrong bowling actions, reduce injury risk, and support evidence-based coaching strategies.

From a theoretical perspective, performing biomechanical analysis with machine learning has introduced new opportunities for understanding and analyzing bowling actions. Data driven methods can capture patterns and relationships between body parts, joint angles and sometimes performance outcomes that traditional methods might miss. By analyzing large amounts of athlete movement data, researchers can identify efficient and safe bowling techniques, helping to prevent injuries and improve performance. These findings directly supports the direction of current research which focuses machine learning based biomechanical evaluation of bowling actions in cricket. With the support of computer vision techniques, it has become much easier to detect skeleton points from vide data and measure key biomechanical parameters like, joint angles, body alignment, stride length, and delivery mechanics without requiring specialized motion capture systems. This direction has the potential to bridge the gap between high-accuracy laboratory systems and practical field-based analysis tools.

In the end, developing a reliable machine learning-based system for analyzing cricket bowling mechanics could provide a low-cost, accurate and field based solution for biomechanical evaluation. This would help coaches and athletes to analyze bowling actions during actual training sessions in order to improve their performances and reduce the risk of injury. Therefore, continuous research in markerless pose estimation and machine learning based biomechanical analysis is really important to make biomechanical evaluation more accurate and widely available in field based sporting environments.

VIII. CONCLUSION

This literature review focuses on achievements and challenges in the use of contemporary biomechanical analysis software in various categories like marker-based motion capture, wearable IMU and markerless approaches. Each category offers specific benefits but also poses disadvantages that restrict its use based on financial resources, accuracy, portability, and ease of use. It indicates the transition from high cost laboratory based motion capture systems to AI/ML motion analysis software. A significant achievement made in the biomechanical analysis AI and ML tools is the development of new technologies. Nonetheless, there remains a substantial deficit in providing valid and dependable frameworks for the accurate and timely examination of movement biomechanics outside a lab that non-specialist users such as coaches, athletes, and trainers, can utilize.

To bridge this gap, future research and development is necessary to develop user friendly AI/ML based solutions that offer accurate and practical analysis for non-technical users. Future studies should also explore cloud-based processing and edge computing to enable real-time biomechanical analysis without the need for complex hardware. Additionally, research

should consider the ethical and privacy factors of AI-driven motion analysis, ensuring that advancements align with responsible data usage and security standards.

Overall, by addressing current challenges, future developments can make biomechanical analysis more effective for a broader range of users, ultimately driving innovation in sports.

REFERENCES

- [1] A. Thander, "Biomechanics of the bowling delivery in cricket: A critical analysis," *International Journal of Novel Research and Development*, vol. 8, no. 6, 2023.
- [2] G. Teferi, "Methods of biomechanical performance analyses in sport: Systematic review," *American Journal of Sports Science and Medicine*, 2020.
- [3] H. U. R. Siddiqui, F. Younas, F. Rustam, E. S. Flores, J. B. Ballester, I. d. I. T. Diez, S. Dudley, and I. Ashraf, "Enhancing cricket performance analysis with human pose estimation and machine learning," *Sensors*, vol. 23, no. 15, p. 6839, 2023.
- [4] G. Alderink and S. Öunpuu, "Biomechanics of human motion and its clinical applications: Instrumented gait analysis," p. 1076, 2025.
- [5] S. L. Colyer, M. Evans, D. P. Cosker, and A. I. Salo, "A review of the evolution of vision-based motion analysis and the integration of advanced computer vision methods towards developing a markerless system," *Sports medicine-open*, vol. 4, no. 1, p. 24, 2018.
- [6] P. Huang, A. Mostovov, R. Cohen, C. Cadilhac, and R. Pionnier, "Comparison of footme insoles with a motion capture system coupled to force plates for assessing gait and posture," *Scientific Reports*, vol. 15, no. 1, p. 13476, 2025.
- [7] G. Nagymáté and R. M. Kiss, "Application of optitrack motion capture systems in human movement analysis: A systematic literature review," *Recent Innovations in Mechatronics*, vol. 5, no. 1, pp. 1–9, 2018.
- [8] J. Vleemeyer, L. Tronicke, L. Schreff, R. Abel, K. Lechler, and R. Müller, "A full-body motion capture gait dataset of healthy young adults walking ramps up and down," *Scientific Data*, 2026.
- [9] M. Al-Amri, K. Nicholas, K. Button, V. Sparkes, L. Sheeran, and J. L. Davies, "Inertial measurement units for clinical movement analysis: Reliability and concurrent validity," *Sensors*, vol. 18, no. 3, p. 719, 2018.
- [10] G. Alderink and S. Öunpuu, "Biomechanics of human motion and its clinical applications: Instrumented gait analysis," p. 1076, 2025.
- [11] J. Taborri, J. Keogh, A. Kos, A. Santuz, R. Abek, K. Urbanczyk, E. van der Kruk, and S. Rossi, "Sport biomechanics applications using inertial, force, and emg sensors: A literature overview," *Applied bionics and biomechanics*, vol. 2020, no. 1, p. 2041549, 2020.
- [12] V. Camomilla, E. Bergamini, S. Fantozzi, and G. Vannozzi, "Trends supporting the in-field use of wearable inertial sensors for sport performance evaluation: A systematic review," *Sensors*, vol. 18, no. 3, p. 873, 2018.
- [13] D. Roetenberg, H. Luinge, P. Slycke *et al.*, "Xsens mvn: Full 6dof human motion tracking using miniature inertial sensors," *Xsens Motion Technologies BV, Tech. Rep.*, vol. 1, no. 2009, pp. 1–7, 2009.
- [14] A. Fleureau, M. Lacome, M. Buchheit, A. Couturier, and G. Rabita, "Validity of an ultra-wideband local positioning system to assess specific movements in handball," *Biology of sport*, vol. 37, no. 4, pp. 351–357, 2020.
- [15] K. J. Boddy, J. A. Marsh, A. Caravan, K. E. Lindley, J. O. Scheffey, and M. E. O'Connell, "Exploring wearable sensors as an alternative to marker-based motion capture in the pitching delivery," *PeerJ*, vol. 7, p. e6365, 2019.
- [16] X. Robert-Lachaine, H. Mecheri, C. Larue, and A. Plamondon, "Validation of inertial measurement units with an optoelectronic system for whole-body motion analysis," *Medical & biological engineering & computing*, vol. 55, no. 4, pp. 609–619, 2017.
- [17] W. Teufel, M. Miezal, B. Taetz, M. Fröhlich, and G. Bleser, "Validity of inertial sensor based 3d joint kinematics of static and dynamic sport and physiotherapy specific movements," *PloS one*, vol. 14, no. 2, p. e0213064, 2019.
- [18] A. Mathis, P. Mamidanna, K. M. Cury, T. Abe, V. N. Murthy, M. W. Mathis, and M. Bethge, "DeepLabCut: markerless pose estimation of user-defined body parts with deep learning," *Nature neuroscience*, vol. 21, no. 9, pp. 1281–1289, 2018.

- [19] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, "Openpose: Realtime multi-person 2d pose estimation using part affinity fields," *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 1, pp. 172–186, 2019.
- [20] —, "Openpose: Realtime multi-person 2d pose estimation using part affinity fields," *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 1, pp. 172–186, 2019.
- [21] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee *et al.*, "Mediapipe: A framework for building perception pipelines," *arXiv preprint arXiv:1906.08172*, 2019.
- [22] S. D. Uhlich, A. Falisse, Ł. Kidziński, J. Muccini, M. Ko, A. S. Chaudhari, J. L. Hicks, and S. L. Delp, "Opencap: Human movement dynamics from smartphone videos," *PLoS computational biology*, vol. 19, no. 10, p. e1011462, 2023.
- [23] A. Toshev and C. Szegedy, "Deeppose: Human pose estimation via deep neural networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 1653–1660.
- [24] B. Van Hooren, N. Pecasse, K. Meijer, and J. M. N. Essers, "The accuracy of markerless motion capture combined with computer vision techniques for measuring running kinematics," *Scandinavian Journal of Medicine & Science in Sports*, vol. 33, no. 6, pp. 966–978, 2023.
- [25] F. Lefebvre, I. Rogowski, N. Long, and Y. Blache, "Three-dimensional markerless pose estimation for anatomical landmarks of the shoulder and upper limb," *Journal of Electromyography and Kinesiology*, p. 103067, 2025.
- [26] L. Needham, M. Evans, D. P. Cosker, L. Wade, P. M. McGuigan, J. L. Bilzon, and S. L. Colyer, "The accuracy of several pose estimation methods for 3d joint centre localisation," *Scientific reports*, vol. 11, no. 1, p. 20673, 2021.
- [27] C. Strongman, F. Cavallerio, M. A. Timmis, and A. Morrison, "A scoping review of the validity and reliability of smartphone accelerometers when collecting kinematic gait data," *Sensors*, vol. 23, no. 20, p. 8615, 2023.